

CodeCare AI: From Written Policy to Executable Rules

An explainable coding copilot that turns ICD-10-CM guidelines into auditable, tested rules

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1. The Topic

Payers and providers run on coding content: the ICD-10-CM Official Guidelines, National Correct Coding Initiative (NCCI) edits, and thousands of payer policies (CMS, 2026). Most of it arrives as prose. For every claim, a coder or a claims system must turn that prose into exact decisions. Which code does the note support? Which codes conflict? What documentation is missing? Coders and auditors still apply these rules by hand, and two people can read the same guideline two ways.

The OECD calls the alternative "rules as code," publishing a machine-executable version of a rule next to the human one (Mohun & Roberts, 2020). This report asks how Cotiviti can turn coding policy into rules that software runs, auditors can trace, and teams can update each October.

2. Trends

- **Coding errors still cost billions.** CMS measured a 6.55% Medicare fee-for-service improper payment rate in FY 2025, or \$28.83 billion, down from 7.66% the year before (ACDIS, 2026).
- **Providers see more denials.** In a 2025 survey of 250 revenue leaders, 41% reported denial rates of 10% or higher and 54% said claim errors are rising. Missing or inaccurate data drove half of all denials (Experian Health, 2025).
- **Providers want AI, but few use it.** 67% believe AI can improve claims, yet 14% use it today. Among those users, 69% report fewer denials (Experian Health, 2025).
- **General LLMs make poor coders on their own.** Mount Sinai researchers tested GPT-4, GPT-3.5, Gemini Pro, and Llama-2-70B. The best model matched 33.9% of ICD-10-CM codes exactly and often returned plausible but wrong codes (Soroush et al., 2024).
- **Vendors are racing to automate coding and rule writing.** KLAS launched a 2026 category for autonomous coding (OmniMD, 2026), Arintra raised \$25 million in August 2026 (Pennic, 2026), and Cotiviti added AI-assisted rule authoring to Payment Policy Management (Cotiviti, 2026).
- **Content changes on a fixed calendar.** ICD-10-CM updates every October 1, CPT every January 1, and NCCI every quarter. Each release can break rules written for the last one (CMS, 2026, n.d.).

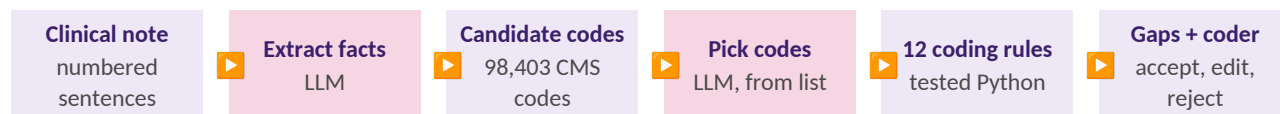
3. Opportunities and Threats

Opportunities	Threats
Speed. LLMs read clinical text and draft rule logic from policy language, cutting the time from new policy to live edit.	Regulators limit automated decisions. CMS told Medicare Advantage plans that coverage decisions must rest on each patient's circumstances, not an algorithm alone (American Hospital Association, 2024). California's SB 1120 bars AI from denying, delaying, or modifying care on its own (Harned, 2024). Tools that act without human review carry legal risk.
Evidence. Linking each code or edit to a chart sentence and a policy section makes findings easier to defend.	Provider AI raises coding intensity. As providers adopt autonomous coding, claims arrive with more codes and higher specificity than manual review can keep up with. Cotiviti names AI-driven provider billing as an emerging payment integrity risk (Cotiviti, n.d.).
Prevention. Flagging missing documentation before billing stops denials instead of recovering money later.	Drift. A silent model or prompt update can change results unnoticed. CodeCare AI records the model, prompt, and code set on every run and replays saved outputs to catch changes.
Safer updates. Versioned rules with regression tests make each October release predictable.	Data limits. PHI rules and AMA-licensed CPT text restrict what data a model can use. CodeCare AI uses synthetic notes and public CMS data only.

4. Proof of Concept: CodeCare AI

To test these ideas, I built CodeCare AI, a working prototype that codes synthetic outpatient notes for type 2 diabetes, chronic kidney disease (CKD), hypertension, and heart failure (Bobba, 2026). The design splits the work in two. **The LLM reads, and plain code checks.** The LLM extracts each condition with the sentence that supports it, then picks

codes from a candidate list drawn from the 98,403 codes in the official FY 2027 CMS tables, so it cannot invent a code. Twelve rules, written as tested Python functions that cite their guideline sections, then fix combination codes and flag gaps.



For a note reading "type 2 diabetes and chronic kidney disease stage 3," the rules return E11.22 plus N18.30 and ask the doctor to document stage 3a or 3b. A coder accepts, edits, or rejects each code on a screen that highlights the evidence. I built it with FastAPI, PostgreSQL with pgvector, LangGraph, the open-weight gpt-oss-120b model on Groq, and Next.js, and compared it with the same model asked for codes directly.

Metric (20 synthetic gold notes, same model)	LLM only	CodeCare AI
Precision: suggested codes that are correct	0.56	0.95
Recall: correct codes the system found	0.49	0.95
Invalid codes: non-existent or non-billable	12%	0%
Documentation gaps flagged	0%	100%

Synthetic notes only. I checked expected codes against the FY 2027 guidelines. A certified coder has not reviewed them.

5. Recommendations for Cotiviti

1. **Pair every AI-drafted rule with tests.** Extend AI-assisted rule authoring in Payment Policy Management so each drafted rule ships with test claims built from the policy text, and require a pass on a gold claim set before an analyst approves it. The LLM drafts, and deterministic code runs in production.
2. **Put the evidence in every finding.** Show the chart sentence and the guideline or policy citation behind each Coding Validation and Clinical Chart Validation finding. Evidence gives providers a reason to accept a finding and gives auditors a faster answer on appeal.
3. **Automate the October 1 update.** When CMS publishes a new code set or NCCI release, diff it against current content, list every affected rule, and rerun its regression tests before the effective date.
4. **Measure AI before trusting it.** Require zero invalid codes, an evidence check, and a head-to-head comparison with an LLM-only baseline before any model reaches production. Coders and auditors keep the final decision.

Autonomous coding startups sell automation rates. Cotiviti's payer clients need decisions they can defend in an audit, and explainable, tested rules give them that. I recommend Cotiviti position itself there.

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